

WMDR2-core

Package in package 'Model'

WMDR2-core

Version 0.2.3 Phase 1.0 Proposed

jkl created on 2026-03-06. Last modified 2026-06-25

WMDR2-core diagram

Class diagram in package 'WMDR2-core'

wmdr2-core v0.2.x

v0.2.0

- time (a temporal extent) shall be realized as time:interval (unchanged from v.0.1.13)
- Following the OGC Features model, geometry = geolocation = coordinates + elevation
- All temporal<Name> elements are renamed to historical<Name>.
- historical<Name> elements other than historicalLocation shall be realized as arrays of dictionary objects for (begin) date, and the values of <Name>, e.g.,

```
historicalTerritory = [{
  "date": "2000-01-01",
  "territory": "<ISO1>"
},
{
  "date": "2024-07-13",
  "territory": "<ISO2>"
}]
```

- "observation" describes time-series (collections) of (individual) observations / measurements.
- "observation"s are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- "deployment" is removed as a concept, most attributes are directly under "observation". - "instrument" links directly to "observation". In alignment with OMS/STA, there is a link to "facility" via "deplyoment", but this is not expected to be used at this point.
- "observation"s can be grouped to an "observationSets" (NB: This is a test and not guaranteed to exist in future.)

v0.2.1

- "observation" has been renamed to "observations", because it describes collections (=time-series) of (individual) observations / measurements.
- "observations" are acquired using a "historicalObservingSchedule", which has a (begin) date and refers to a schedule / coverage.
- The concept of deployment has been developed into "historicalDeployment", which is now a more focussed and intuitive concept. It describes the placement at a specific date of an instrument at a location (=geometry) that can generate observations. It describes the "exposure" of the instrument as well as its "operatingStatus" and "serialNumber" from that date onwards.
- "observations" can be grouped to an "observationSets" (NB: This is experimental and not guaranteed to exist in future.)
- "reporting" and "historicalReporting" are modeled to allow reuse of a reporting template. This is similar to the "schedule" / "historicalObservingSchedule" concept.

v0.2.2

- clean up attribute names, consistent singular naming within historical<Name> classes,

v0.2.3

- Class names capitalized, except if class is re-usable.
- prefix "historical" removed as not needed (TT-WMDR2, 2026-06-24), the "date" element makes this clear
- "observations" renamed to "ObservationSeries" to avoid mis-interpretation.
- update cardinalities on instrument.manufacturer to [0..1], instrument.model to [0..1]
- remove ObservationSeries.time (can be inferred from Deployments.)
- rename ObservingSchedule to ObservingProcedure and add "strategy", rename schedule to observingSchedule, rename sampling to samplingFrequency, aggregation to aggregationInterval
- rename Reporting to ReportingProcedure and add strategy; remove aggregation (a misnomer and different concept, used as an alias of temporalReportingInterval), add ReportingProcedure.reportingSchedule to allow regular but

WMDR2-core
Version 0.2.1
2026-06-25

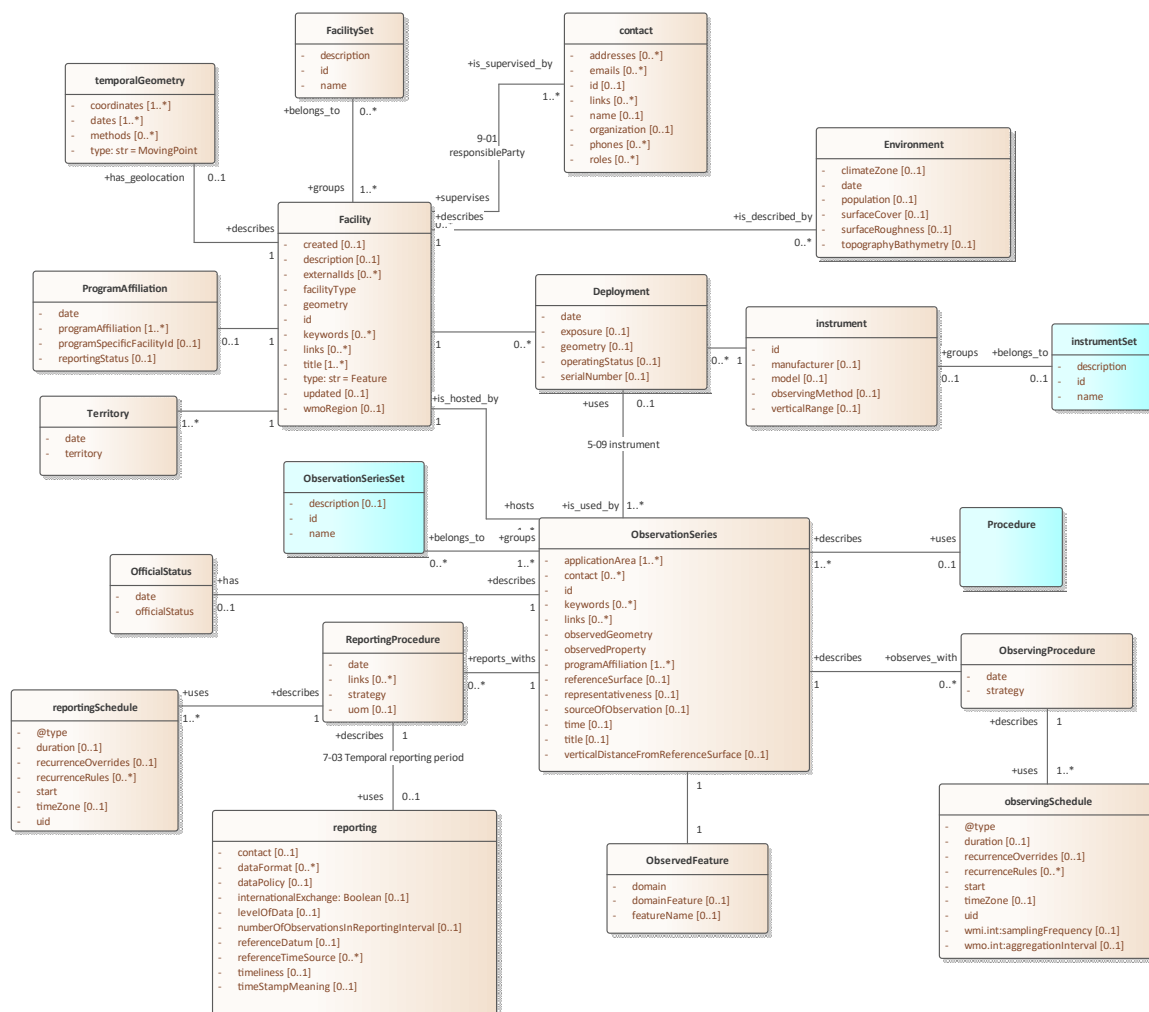


Figure 1: WMDR2-core

Class in package 'WMDR2-core'

WMDR2 core JSON class: contact.

contact
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES	
	addresses : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES	
Postal addresses.	[Is static True. Containment is Not Specified.]
 emails : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Email contacts.	[Is static True. Containment is Not Specified.]
 id : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Optional stable contact identifier.	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Online resources associated with the contact.	[Is static True. Containment is Not Specified.]
 name : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Either a full name, or a sequence of name parts. Examples: Dr. Jörg Klausen (this corresponds to FN in vcard) Klausen;Jörg;Dr.; (this corresponds to N in vcard)	[Is static True. Containment is Not Specified.]
 organization : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Example: ORG:Federal Office of Meteorology and Climatology MeteoSwiss	[Is static True. Containment is Not Specified.]
 phones : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Phone contacts.	[Is static True. Containment is Not Specified.]
 roles : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Specific contact roles such as owner, pointOfContact, principalInvestigator.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	

ASSOCIATIONS

 Association (direction: Unspecified) 9-01 responsibleParty

Source: Public supervises (Class) Facility
Cardinality: [0..*]

Target: Public is_supervised_by (Class) contact
Cardinality: [1..*]

Deployment

Class in package 'WMDR2-core'

Deployment

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-25

ATTRIBUTES

 date : Private

[Is static True. Containment is Not Specified.]

 exposure : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/Exposure>

[Is static True. Containment is Not Specified.]

 geometry : Private

Alias: temporalGeometry

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

geometry is the OGC Records name for geolocation.

[Is static True. Containment is Not Specified.]

 operatingStatus : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

values refer to <http://codes.wmo.int/wmdr/InstrumentOperatingStatus>

[Is static True. Containment is Not Specified.]

 serialNumber : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) Deployment
Cardinality: [0..*]

Target: Public (Class) instrument
Cardinality: [1]

 Association (direction: Unspecified)

ASSOCIATIONS	
Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) Deployment Cardinality: [0..*]
 Association (direction: Unspecified) 5-09 instrument	
Source: Public is_used_by (Class) ObservationSeries Cardinality: [1..*]	Target: Public uses (Class) Deployment Cardinality: [0..1]

Environment

Class in package 'WMDR2-core'

Facility-environment object containing climate, surface-cover, population, and topography/bathymetry histories.

Environment
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-02. Last modified 2026-06-25

ATTRIBUTES	
 climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 date : Private	[Is static True. Containment is Not Specified.]
 population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility	[Is static True. Containment is Not Specified.]
 surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	

ASSOCIATIONS

Source: Public describes (Class) Facility
Cardinality: [1]

Target: Public is_described_by (Class)
Environment
Cardinality: [0..*]

Facility

Class in package 'WMDR2-core'

Facility properties payload of the WMDR2 full record. Root Features geometry/time are modeled on fullRecord, not here.

Facility
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES

 created : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Record creation date/time metadata.

[Is static True. Containment is Not Specified.]

 description : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Facility description.

[Is static True. Containment is Not Specified.]

 externalIds : Private
Alias: 3-06 (primary & additional) WSIs
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

External identifiers such as WMO:WIGOS.

[Is static True. Containment is Not Specified.]

 facilityType : Private
Alias: 3-04 facilityType

Compact WMDR facility type code value.

[Is static True. Containment is Not Specified.]

 geometry : Private
Alias: 3-07 current geolocation

geometry = geolocation = coordinates + elevation

[Is static True. Containment is Not Specified.]

 id : Private
Alias: 3-06 (primary) WSI

to prefixed with name space "facility:"

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
🔑 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON.	[Is static True. Containment is Not Specified.]
🔑 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static True. Containment is Not Specified.]
🔑 title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	[Is static True. Containment is Not Specified.]
🔑 type : str Private = Feature Facility properties type; const facility.	[Is static True. Containment is Not Specified.]
🔑 updated : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Record update date/time metadata.	[Is static True. Containment is Not Specified.]
🔑 wmoRegion : Private Alias: 3-01 wmoRegion Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) mandatory for fixed facilities, optional for mobile facilities.	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
✎ Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1] Target: Public (Class) Deployment Cardinality: [0..*]	
✎ Association (direction: Unspecified) Source: Public is_hosted_by (Class) Facility Cardinality: [1] Target: Public hosts (Class) ObservationSeries Cardinality: [1..*]	

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) Territory Cardinality: [1..*]
 Association (direction: Unspecified) Source: Public groups (Class) Facility Cardinality: [1..*]	Target: Public belongs_to (Class) FacilitySet Cardinality: [0..*]
 Association (direction: Unspecified) Source: Public describes (Class) Facility Cardinality: [1]	Target: Public is_described_by (Class) Environment Cardinality: [0..*]
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]
 Association (direction: Unspecified) Source: Public describes (Class) Facility Cardinality: [1]	Target: Public has_geolocation (Class) temporalGeometry Cardinality: [0..1]
 Association (direction: Unspecified) 9-01 responsibleParty Source: Public supervises (Class) Facility Cardinality: [0..*]	Target: Public is_supervised_by (Class) contact Cardinality: [1..*]

FacilitySet

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

FacilitySet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 description : Private	[Is static True. Containment is Not Specified.]
 id : Private	[Is static True. Containment is Not Specified.]

ATTRIBUTES

◆ name : Private
Alias: title

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

✎ Association (direction: Unspecified)

Source: Public groups (Class) Facility
Cardinality: [1..*]

Target: Public belongs_to (Class) FacilitySet
Cardinality: [0..*]

instrument

Class in package 'WMDR2-core'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like observedProperty and observedGeometry are used, even though they just describe the instrument capabilities, not the actual observations. With this, instrument becomes re-usable between deployments and even facilities.

instrument

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-05-22. Last modified 2026-06-25

ATTRIBUTES

◆ id : Private

[Is static True. Containment is Not Specified.]

◆ manufacturer : Private
Alias: 5-09 manufacturer
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

◆ model : Private
Alias: 5-09 model
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]

◆ observingMethod : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

from (new) unified code list of observing methods, based on
<http://codes.wmo.int/wmdr/ObservingMethodAtmosphere>
<http://codes.wmo.int/wmdr/ObservingMethodTerrestrial>
<http://codes.wmo.int/wmdr/ObservingMethodEarth>

[Is static True. Containment is Not Specified.]

◆ verticalRange : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

ATTRIBUTES

Specify using an array of [lower boundary, upper boundary] in meter. This is a capability of an instrument as specified, not the actually accessible vertical range.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public (Class) Deployment
Cardinality: [0..*]

Target: Public (Class) instrument
Cardinality: [1]

instrumentSet

Class in package 'WMDR2-core'

instrumentSet

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-22. Last modified 2026-06-25

ATTRIBUTES

 description : Private

[Is static True. Containment is Not Specified.]

 id : Private

[Is static True. Containment is Not Specified.]

 name : Private
Alias: title

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public groups (Class) instrument
Cardinality: [0..1]

Target: Public belongs_to (Class) instrumentSet
Cardinality: [0..1]

ObservationSeries

Class in package 'WMDR2-core'

WMDR2 core JSON class: observation.

ObservationSeries
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
◆ applicationArea : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Value(s) from http://codes.wmo.int/wmdr/ApplicationArea [Is static True. Containment is Not Specified.]	
◆ contact : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation-level contacts. [Is static True. Containment is Not Specified.]	
◆ id : Private Observation identifier, e.g. observation:<observedVariable>. [Is static True. Containment is Not Specified.]	
◆ keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation keywords. [Is static True. Containment is Not Specified.]	
◆ links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Observation links. [Is static True. Containment is Not Specified.]	
◆ observedGeometry : Private cf. http://codes.wmo.int/wmdr/Geometry [Is static True. Containment is Not Specified.]	
◆ observedProperty : Private Alias: OMS:ObservedProperty, iop:Variable The observedVariable is constructed following I-ADOPT ontology approaches, with elements described in a unified vocabulary hosted at codes.wmo.int/wmdr2. NB: observedVariable will be a LONG list of entries, each record comprised of notation, name, description. The notation is a unique identifier, not necessarily human-readable. The name is potentially human-readable. The description is a fully fledged definition, human-readable and describing all relevant aspects of an observed variable. It is derived from the observedProperty, the object of interest, the matrix and domain, as well as constraints. Matrix and constraints are defined in controlled vocabularies that may be domain-specific. Examples include: - observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ...), or another property (e.g., thickness, concentration, scattering coefficient, color, ...) - objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll	

ATTRIBUTES

- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground for temperature measurements), a size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature (range), a humidity (range), etc., as relevant for the result of the observation. Constraints shall be documented as name-value pairs, where the names shall come from a code list, and the values shall come from a code list depending on the name, or be free text.
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, dry air, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

Applications may want to guide users to select a specific observedVariable by offering groupings, filtering, or tree-like visualizations according to matrix, objectOfInterest, etc.

[Is static True. Containment is Not Specified.]

 programAffiliation : Private
Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Program-affiliation timeline fragments inherited from WMDR1 observations.

[Is static True. Containment is Not Specified.]

 referenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Linked to heightAboveReferenceSurface, from <http://codes.wmo.int/wmdr/ReferenceSurfaceType>

[Is static True. Containment is Not Specified.]

 representativeness : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

<http://codes.wmo.int/wmdr/Representativeness>

[Is static True. Containment is Not Specified.]

 sourceOfObservation : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/SourceOfObservation>.

If sourceOfObservation=="humanObservation", several attributes of "observations" are expected to be null.

[Is static True. Containment is Not Specified.]

 time : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Observation temporal extent, date-resolution interval or null.

[Is static True. Containment is Not Specified.]

 title : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Human-readable title derived from domain, geometry, observed variable, and optional label.

[Is static True. Containment is Not Specified.]

 verticalDistanceFromReferenceSurface : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

implement as an array of objects:

ATTRIBUTES

[{"value": float, "uom": str}, {...}], whereby "uom" from <http://codes.wmo.int/wmdr/unit>
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public reports_with (Class)
ReportingProcedure
Cardinality: [0..*]

 Association (direction: Unspecified) 5-09 instrument

Source: Public is_used_by (Class) ObservationSeries
Cardinality: [1..*]

Target: Public uses (Class) Deployment
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public (Class) ObservedFeature
Cardinality: [1]

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1..*]

Target: Public uses (Class) Procedure
Cardinality: [0..1]

ObservingProcedure(s)

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public has (Class) OfficialStatus
Cardinality: [0..1]

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public observes_with (Class)
ObservingProcedure
Cardinality: [0..*]

 Association (direction: Unspecified)

Source: Public groups (Class) ObservationSeries
Cardinality: [1..*]

Target: Public belongs_to (Class)
ObservationSeriesSet
Cardinality: [0..*]

 Association (direction: Unspecified) 3-07 temporalGeometry

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public (Class) temporalGeometry
Cardinality: [0..1]

ASSOCIATIONS	
describes	has_geolocatio
 Association (direction: Unspecified) Source: Public is_hosted_by (Class) Facility Cardinality: [1] Target: Public hosts (Class) ObservationSeries Cardinality: [1..*]	

ObservationSeriesSet

Class in package 'WMDR2-core'

grouping of observations (which are themselves time series)

ObservationSeriesSet
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-06. Last modified 2026-06-25

ATTRIBUTES	
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Deployment temporal extent, date-resolution interval or null.	[Is static True. Containment is Not Specified.]
 id : Private Deployment identifier; source XML id is preserved when available.	[Is static True. Containment is Not Specified.]
 name : Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public groups (Class) ObservationSeries Cardinality: [1..*] Target: Public belongs_to (Class) ObservationSeriesSet Cardinality: [0..*]	

ObservedFeature

Class in package 'WMDR2-core'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-feature-of-interest, of an observation.

ObservedFeature
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-10. Last modified 2026-06-25

ATTRIBUTES

domain : Private

The largest scale in WIGOS, cf. <http://codes.wmo.int/wmdr/Domain>

[Is static True. Containment is Not Specified.]

domainFeature : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the type of the larger-scale environmental target or ultimate-feature-of-interest, of an observation.

Concept motivated by requirements from hydrology, but may be useful elsewhere. New code list is proposed, with terms like

- river basin
- lake
- river
- catchment
- canopy
- epilimnion
- hypolimnion
- troposphere
- tropopause
- stratosphere

- thermocline (useful as a general term, but needs "domain")
- boundary layer (useful as a general term, but needs "domain")

- ...

[Is static True. Containment is Not Specified.]

featureName : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

free text field, e.g., "Danube Basin", "Amazon rain forest", "Amazon river Basin"

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public (Class) ObservedFeature
Cardinality: [1]

ObservingProcedure

Class in package 'WMDR2-core'

When are observations made? Describes an observingStrategy, and depending on it, a temporal schedule/coverage, e.g. "from Feb-Oct on working days from 06-18 UTC".

Implement as array of objects

ObservingProcedure

Version 1.0 Phase 1.0 Proposed
jkl created on 2026-03-11. Last modified 2026-06-25

ATTRIBUTES	
 date : Private	
Date from when on the schedule is valid	[Is static True. Containment is Not Specified.]
 strategy : Private	
Values from http://codes.wmo.int/wmdr/SamplingStrategy	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservingProcedure Cardinality: [1]	Target: Public uses (Class) observingSchedule Cardinality: [1..*]
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1]	Target: Public observes_with (Class) ObservingProcedure Cardinality: [0..*]

observingSchedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific extensions are represented with wmo.int:* properties.

May be expected for strategy "continuous" or "discrete"

observingSchedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-25

ATTRIBUTES	
 @type : Private	
JSCalendar object type; const Event.	[Is static True. Containment is Not Specified.]
 duration : Private	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Occurrence duration describing coverage window or sampling duration.	[Is static True. Containment is Not Specified.]

ATTRIBUTES

recurrenceOverrides : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence override patches; null override values are invalid.

[Is static True. Containment is Not Specified.]

recurrenceRules : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence rules.

[Is static True. Containment is Not Specified.]

start : Private

Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.

[Is static True. Containment is Not Specified.]

timeZone : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

IANA time zone or UTC default.

[Is static True. Containment is Not Specified.]

uid : Private

JSCalendar-safe schedule uid.

[Is static True. Containment is Not Specified.]

wmi.int:samplingFrequency : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

The frequency at which raw / level 0 observations are obtained.

Implement as {"value": int, "uom": str}

[Is static True. Containment is Not Specified.]

wmo.int:aggregationInterval : Private
 Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the smallest interval over which raw/ level 0 observations are aggregated and stored.

Implement as {"value": int, "uom": str}

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified)

Source: Public describes (Class) ObservingProcedure
 Cardinality: [1]

Target: Public uses (Class) observingSchedule
 Cardinality: [1..*]

OfficialStatus

Class in package 'WMDR2-core'

OfficialStatus

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES

 date : Private

[Is static True. Containment is Not Specified.]

 officialStatus : Private

Values from <http://codes.wmo.int/wmdr/ObservationStatus>

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1]

Target: Public has (Class) OfficialStatus
Cardinality: [0..1]

Procedure

Class in package 'WMDR2-core'

Retained from previous UML model; not part of the current WMDR2 core full-record JSON schema.

Procedure

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-25

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public describes (Class) ObservationSeries
Cardinality: [1..*]

Target: Public uses (Class) Procedure
Cardinality: [0..1]

ObservingProcedure(s)

ProgramAffiliation

Class in package 'WMDR2-core'

ProgramAffiliation

Version 1.0 Phase 1.0 Proposed

ChatGPT created on 2026-06-01. Last modified 2026-06-25

Alias temporalProgramAffiliation

ATTRIBUTES	
 date : Private	[Is static True. Containment is Not Specified.]
 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 programSpecificFacilityId : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Facility Cardinality: [1]	Target: Public (Class) ProgramAffiliation Cardinality: [0..1]

reporting

Class in package 'WMDR2-core'

Reusable reporting template.

reporting
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-01. Last modified 2026-06-23

ATTRIBUTES	
 contact : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static True. Containment is Not Specified.]
 dataFormat : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Aligned array, values from http://codes.wmo.int/wmdr/DataFormat.	[Is static True. Containment is Not Specified.]
 dataPolicy : Private	

ATTRIBUTES

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/DataPolicy>.

[Is static True. Containment is Not Specified.]



internationalExchange : Boolean Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]



levelOfData : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/LevelOfData>.

[Is static True. Containment is Not Specified.]



numberOfObservationsInReportingInterval : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

[Is static True. Containment is Not Specified.]



referenceDatum : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Value from <http://codes.wmo.int/wmdr/CoordinateReferenceSystem>.

[Is static True. Containment is Not Specified.]



referenceTimeSource : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Single value or array, values from <http://codes.wmo.int/wmdr/ReferenceTime>.

[Is static True. Containment is Not Specified.]



timeliness : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Expected timeliness of reporting, an object with value and uom (default: seconds)

[Is static True. Containment is Not Specified.]



timeStampMeaning : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Aligned array, values from <http://codes.wmo.int/wmdr/TimeStampMeaning>.

[Is static True. Containment is Not Specified.]

ASSOCIATIONS

Association (direction: Unspecified) 7-03 Temporal reporting period

Source: Public describes (Class) ReportingProcedure

Cardinality: [1]

Target: Public uses (Class) reporting

Cardinality: [0..1]

ReportingProcedure

Class in package 'WMDR2-core'

ReportingProcedure
Version 1.0 Phase 1.0 Proposed
jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES	
 date : Private	[Is static True. Containment is Not Specified.]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval.	[Is static True. Containment is Not Specified.]
 strategy : Private	[Is static True. Containment is Not Specified.]
 uom : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Values from http://codes.wmo.int/wmdr/unit .	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public describes (Class) ReportingProcedure Cardinality: [1]	Target: Public uses (Class) reportingSchedule Cardinality: [1..*]
 Association (direction: Unspecified) 7-03 Temporal reporting period	
Source: Public describes (Class) ReportingProcedure Cardinality: [1]	Target: Public uses (Class) reporting Cardinality: [0..1]
 Association (direction: Unspecified)	
Source: Public describes (Class) ObservationSeries Cardinality: [1]	Target: Public reports_with (Class) ReportingProcedure Cardinality: [0..*]

reportingSchedule

Class in package 'WMDR2-core'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule references. WMDR-specific

extensions are represented with wmo.int:* properties.

reportingSchedule
Version 1.0 Phase 1.0 Proposed
ChatGPT created on 2026-06-25. Last modified 2026-06-25

ATTRIBUTES	
💎 @type : Private JSCalendar object type; const Event. [Is static True. Containment is Not Specified.]	
💎 duration : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Occurrence duration describing coverage window or sampling duration. [Is static True. Containment is Not Specified.]	
💎 recurrenceOverrides : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence override patches; null override values are invalid. [Is static True. Containment is Not Specified.]	
💎 recurrenceRules : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) JSCalendar recurrence rules. To be used for "reportingInterval"; use start, duration and other elements to evaluate constraints if needed. [Is static True. Containment is Not Specified.]	
💎 start : Private Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start. [Is static True. Containment is Not Specified.]	
💎 timeZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) IANA time zone or UTC default. [Is static True. Containment is Not Specified.]	
💎 uid : Private JSCalendar-safe schedule uid. [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
🔗 Association (direction: Unspecified) Source: Public describes (Class) ReportingProcedure Target: Public uses (Class) reportingSchedule	

ASSOCIATIONS

Cardinality: [1]

Cardinality: [1..*]

temporalGeometry

Class in package 'WMDR2-core'

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-03-11. Last modified 2026-06-25

Alias temporalGeometry

ATTRIBUTES



coordinates : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static True. Containment is Not Specified.]



dates : Private

Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static True. Containment is Not Specified.]



methods : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static True. Containment is Not Specified.]



type : str Private = MovingPoint

[Is static True. Containment is Not Specified.]

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public describes (Class) Facility

Cardinality: [1]

Target: Public has_geolocation (Class)

temporalGeometry

Cardinality: [0..1]

temporalGeometry

Class in package 'WMDR2-core'

WMDR2 core JSON class: temporalGeometry.

temporalGeometry

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-12. Last modified 2026-06-02

ATTRIBUTES	
 coordinates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Coordinate history array; each entry is lon, lat[, elevation]. [Is static True. Containment is Not Specified.]	
 dates : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) Date array aligned with coordinates. [Is static True. Containment is Not Specified.]	
 methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static True. Containment is Not Specified.]	
ASSOCIATIONS	
 Association (direction: Unspecified) 3-07 temporalGeometry Source: Public (Class) ObservationSeries Cardinality: [1] describes	Target: Public (Class) temporalGeometry Cardinality: [0..1] has_geolocatio

Territory

Class in package 'WMDR2-core'

WMDS reference: 3-02

mandatory for fixed facilities, optional for mobile facilities.

Implemented as [{"date": date, "territory": str}, {}] like other historical<Name> classes.

Territory

Version 1.0 Phase 1.0 Proposed

jkl created on 2026-06-23. Last modified 2026-06-25

ATTRIBUTES	
 date : Private [Is static True. Containment is Not Specified.]	
 territory : Private Alias: temporal Territory Values from http://codes.wmo.int/wmdr/TerritoryName [Is static True. Containment is Not Specified.]	

ASSOCIATIONS Association (direction: Unspecified)Source: Public (Class) Facility
Cardinality: [1]Target: Public (Class) Territory
Cardinality: [1..*]